

Project 2.6

Smart Light-Activated Relay System with Arduino

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Course	Electronics & Embedded Systems
Project Level	2.6 — Individual Portfolio Work
Simulation Tool	Wokwi (Arduino Uno)
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1. Introduction

This document presents Project 2.6, an individual extension of Project 2.5 — the Power Driver circuit using a Relay and Flyback Diode. The objective of Project 2.6 is to elevate the existing analog design by introducing an Arduino Uno microcontroller, converting the circuit from a passive analog switch into a programmable, intelligent embedded system.

The fundamental concept is preserved: a Light-Dependent Resistor (LDR) detects ambient light and triggers a high-power load (lamp) through a relay. The upgrade lies in how this detection is processed — instead of a transistor threshold determined by fixed resistor values, the Arduino reads the LDR signal digitally via its Analog-to-Digital Converter (ADC), applies a software-defined threshold, and simultaneously controls both the relay and an LED status indicator. This represents the real-world transition from analog to digital embedded control.

2. Circuit Description

The circuit is composed of eight components, each serving a specific and deliberate role within the system:

Component	Specification	Function
LDR (Photoresistor)	Light-dependent	Senses ambient light; resistance rises in darkness, falls in light
Resistor	10 kΩ	Forms a voltage divider with the LDR; outputs analog signal to Arduino A0

Arduino Uno	ATmega328P MCU	Reads LDR via ADC, executes threshold logic, drives output pins
Relay Module	5V SPDT	Electrically isolates and switches the high-power lamp load safely
Flyback Diode	1N4148	Suppresses inductive voltage spike when relay coil de-energises
LED	Red, ~2V V_f	Visual status indicator — illuminates when relay is active
Resistor	220 Ω	Current-limiting resistor protecting the LED from overcurrent
Lamp	AC load	High-power output controlled through the relay's switching contacts

3. Circuit Schematic & Simulation

The circuit was designed and validated in Wokwi, an online Arduino simulation environment. The diagram below shows the complete wiring as implemented:

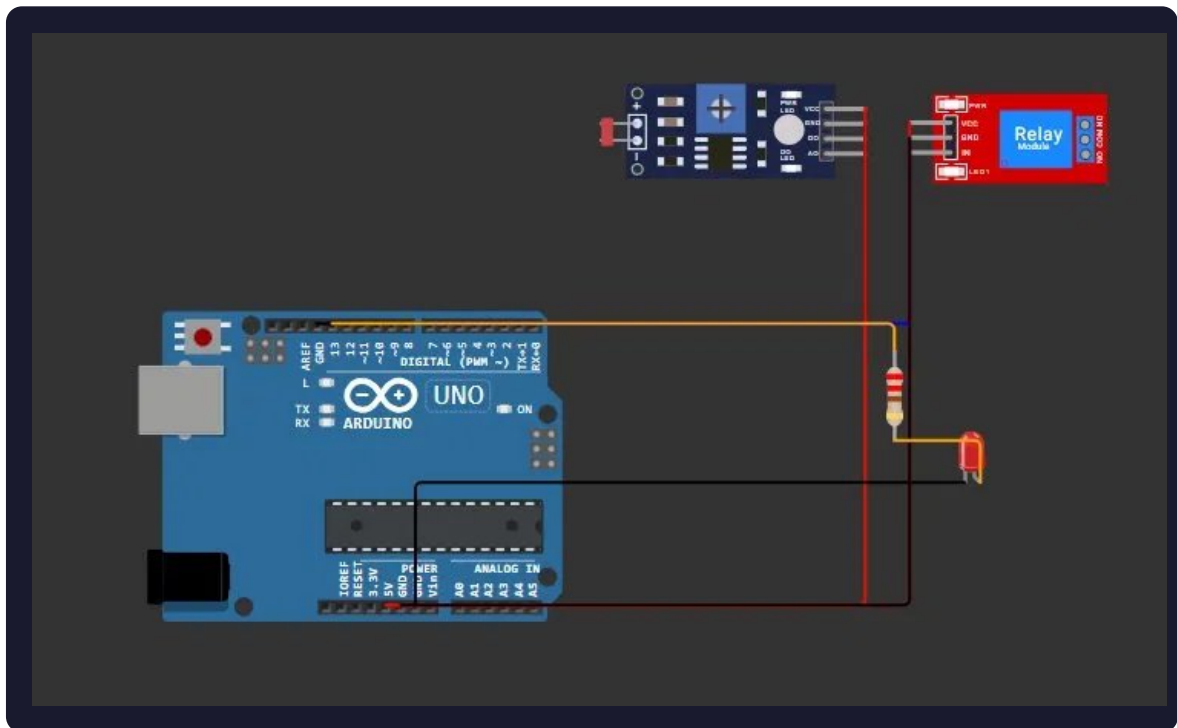


Figure 1 — Wokwi simulation: Arduino Uno with LDR sensor module, relay module, LED, and lamp load.

Wiring connections are summarised as follows:

- LDR Sensor Module AO pin → Arduino A0 | VCC → 5V | GND → GND
- Arduino Pin 7 → Relay Module IN (trigger signal)
- Arduino Pin 13 → 220 Ω resistor → LED anode; LED cathode → GND
- Relay COM / NO terminals → Lamp load (switched output)
- 1N4148 Flyback Diode → across relay coil (cathode toward VCC rail)

4. Code Explanation

The Arduino sketch below implements the full control logic. Variables are declared at the top for easy adjustment; the loop reads the LDR every 100 ms and updates both outputs based on a configurable threshold:

```
int ldrPin = A0; // LDR analog input
int relayPin = 7; // Relay output pin
int ledPin = 13; // LED indicator pin
int threshold = 500; // Midpoint of 0-1023 ADC range

void setup() {
  pinMode(relayPin, OUTPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600); // Serial monitor for debugging
}

void loop() {
  int ldrValue = analogRead(ldrPin); // Read light level (0-1023)
  Serial.println(ldrValue); // Print for monitoring
  if (ldrValue < threshold) { // DARK: value below threshold
    digitalWrite(relayPin, HIGH); // Activate relay
    digitalWrite(ledPin, HIGH); // Turn LED on
  } else { // BRIGHT: value above threshold
    digitalWrite(relayPin, LOW); // Deactivate relay
    digitalWrite(ledPin, LOW); // Turn LED off
  }
  delay(100); // 100 ms polling interval
}
```

The threshold of 500 represents the midpoint of the Arduino's 10-bit ADC (range 0–1023). In darkness the LDR resistance increases, pulling the A0 voltage low — below 500 — which activates both outputs. In bright conditions the voltage rises above 500 and both outputs deactivate. The threshold can be tuned in software without any hardware modification, which is a key advantage over the analog design in Project 2.5.

5. Upgrade from Project 2.5

Project 2.5 implemented a purely analog relay driver: an LDR and resistor formed a voltage divider that biased a BJT transistor into saturation, which then energised the relay coil. While this achieved the goal of switching a high-power load, the design had a fixed, component-dependent threshold with no programmability or status feedback.

Upgrade	Project 2.5	Project 2.6
Control Method	Analog transistor bias	Digital Arduino ADC + software logic
Threshold	Fixed by resistor values	Software variable — adjustable anytime

Status Feedback	None	LED indicator on Pin 13
Expandability	Limited	Serial monitor, IoT, sensor fusion ready
Complexity Level	Analog passive	Embedded microcontroller system

6. Conclusion and Reflection

Project 2.6 successfully demonstrates the transition from a passive analog relay circuit to a microcontroller-driven embedded system. By integrating the Arduino Uno, the design gains programmability, visual feedback, real-time serial monitoring, and a clear path toward further expansion — all while preserving the core safety features of the original design (flyback diode protection, relay isolation).

This project reinforced practical understanding of analog-to-digital conversion, threshold-based conditional logic, relay driving from a microcontroller output pin, and simulation-based prototyping. The use of AI-assisted tools for circuit design and code generation was applied critically — validating outputs against course concepts rather than accepting them blindly — reflecting the modern engineering mindset encouraged by the course.

All project materials — simulation file, source code, this document, and the video explanation — are compiled in my GitHub portfolio and published on Academia.edu as required by the submission guidelines.